

Data Structures and Algorithms - Assignment 1

1. Write all possibilities to check palindrome(number, array, string, linked list, tree) on paper. Also do time and space complexity analysis of all and compare.
2. Write a function to search a Node of given key into singly linked list. (stop search when key is found)
3. In singly linear linked list implement following operations.
 - i. insert a new node after a given node(data)
 - ii. insert a new node before a given node(data)
4. Write a function which will insert values in sorted order into linked list.
5. Implement singly circular linked list using tail pointer (no head pointer). Compare time complexities of all operations.